Depth-Aware Neural Style Transfer

CS 445 Final Project Report

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Abstract

Conventional neural style transfer applies a shared style objective across an entire image without explicitly accounting for depth. While this approach creates stylized images, it can obscure foreground details and make the scene appear flat. Our project introduces depth awareness to a traditional CNN-based style transfer pipeline by combining the Gatys method with MiDaS monocular depth estimation. We separate each content image into foreground, midground, and background using soft depth masks, then blend a single stylized result with the original image at strengths of 30%, 70%, and 100%, respectively. We evaluate the method on 100 content images and five styles, producing 1,000 outputs across the baseline and depth-aware variants. The reported mean SSIM increases from 0.2250 to 0.4087, while PSNR increases from 10.5144 to 13.6209 dB. Perceptual and pixel-space metrics also indicate that the depth-aware outputs remain closer to the original content. These results support depth-dependent blending as a practical way to preserve foreground detail while retaining stronger stylization in distant regions.

CCS Concepts

• **Computing methodologies** → **Image manipulation**; *Neural networks.*

Keywords

neural style transfer, monocular depth estimation, depth-aware blending, computational photography

1 Motivation and Impact

Conventional neural style transfer uses a convolutional neural network (CNN) to combine the content of one image with the artistic style of another. The Gatys method [1] applies a common content and style objective across the image, without explicitly distinguishing objects at different depths. While this approach works well for creating stylized images, the output can appear flat when strong textures obscure the structure of foreground subjects.

In natural photography and classical art, the level of detail and emphasis often varies with depth. Foreground subjects can have sharper detail and less abstraction, while backgrounds appear softer or more heavily stylized. Our project uses this observation to vary stylization across the foreground, midground, and background. By integrating depth estimation with neural style transfer, we aim to preserve the spatial structure of the original image while giving distant regions a stronger artistic effect.

Through this project, we aimed to deepen our understanding of the intersection between neural image generation, geometry, and computational photography aesthetics. Beyond our academic interest, this approach could help photo editing software apply stylization selectively based on scene depth. In augmented and mixed reality, it could also help adjust the abstraction of virtual objects to match surrounding regions. These are potential applications of the method; our experiments focus on still-image stylization.

Our main contribution is a depth-based compositing strategy built around a single baseline stylization. We also developed an interactive tool for refining depth masks and an evaluation pipeline that combines structural, perceptual, and style-content diagnostics. Together, these components let us examine how depth-dependent blending changes the balance between content preservation and artistic abstraction.

2 Approach

2.1 Baseline Neural Style Transfer

We implemented the Gatys style transfer method [1] using a pretrained VGG19 network with ImageNet weights as the feature extractor. We extract content representations from `conv4_2` and style representations from `conv1_1`, `conv2_1`, `conv3_1`, `conv4_1`, and `conv5_1`. We replace max pooling with average pooling for smoother gradients.

Let C be the content image, S the style image, and X the image being optimized. If $F_l(\cdot)$ denotes the features at layer l and $G_l(\cdot)$ their Gram matrix, the objective can be written as

$$\mathcal{L}_{\text{content}}(X, C) = \text{MSE}\big(F_c(X), F_c(C)\big), \tag{1}$$

$$\mathcal{L}_{\text{style}}(X, S) = \sum_{l \in \mathcal{S}} w_l \ \text{MSE}\big(G_l(X), G_l(S)\big), \tag{2}$$

$$\mathcal{L}_{\text{total}} = \lambda_c \mathcal{L}_{\text{content}} + \lambda_s \mathcal{L}_{\text{style}}. \tag{3}$$

Here, c is the content layer, $\mathcal{S}$ is the set of style layers, and the weights control their contributions. We use L-BFGS to optimize the output image while keeping the feature extractor fixed. The resulting stylized image, denoted T, serves both as our baseline and as the input to depth-aware blending.

2.2 Depth Estimation and Spatial Layers

We used MiDaS v3 [2] for monocular depth estimation. Each content image passes through the pretrained model to produce a relative depth map in which higher values indicate closer objects. We normalize these values to $[0, 1]$ for consistent processing. This map provides relative ordering within the scene, rather than physical distances.

Using the normalized depth map d, we separate the image into three spatial layers. Our initial thresholds assign pixels with $d \geq 0.6$ to the foreground, $0.3 < d < 0.6$ to the midground, and $d \leq 0.3$ to the background. These thresholds are starting points: we adjust them interactively through a GUI when they do not separate the scene appropriately.

Hard depth masks can create visible edges between regions. To reduce these artifacts, we apply Gaussian feathering with a kernel size of 31. We then normalize the three masks so their weights sum to one at every pixel. If $\widetilde{M}_k$ is the blurred mask for layer k, the

normalized mask is

$$M_k(p) = \frac{\widetilde{M}_k(p)}{\sum_{j\in\{f,m,b\}} \widetilde{M}_j(p)}. \tag{4}$$

Here, p denotes a pixel. The layer indices f, m, and b refer to foreground, midground, and background. The normalized weights satisfy $\sum_k M_k(p) = 1$, supporting smooth alpha blending across layer boundaries.

2.3 Depth-Dependent Blending

We implemented a variable blending approach to balance visual quality and computational cost. We apply style transfer once to the entire image, then blend the stylized result with the original content using layer-specific weights. We use 30% stylization in the foreground to preserve details, 70% in the midground to provide a transition, and 100% in the background to retain the full stylized effect. These values were selected through empirical tuning across the project images.

With $\alpha_f = 0.3$, $\alpha_m = 0.7$, and $\alpha_b = 1.0$, the final output is

$$Y(p) = \sum_{k\in\{f,m,b\}} M_k(p)\big[\alpha_k T(p) + (1-\alpha_k)C(p)\big]. \tag{5}$$

Equivalently, we can define a spatial strength map

$$A(p) = 0.3M_f(p) + 0.7M_m(p) + M_b(p) \tag{6}$$

and write $Y(p) = A(p)T(p) + (1 - A(p))C(p)$. This formulation makes the role of depth explicit: depth controls how much of the stylized result appears at each pixel. It does not change the VGG feature extractor or introduce a depth loss into the optimization.

This design avoids running style transfer separately for each depth layer. After generating T, we can adjust the masks or blending strengths without repeating the baseline optimization. The soft masks produce gradual transitions while keeping foreground regions closer to the original content.

3 Experimental Setup

3.1 Images and Outputs

We evaluate the depth-aware method against the baseline across 100 distinct content images and five styles. This gives 500 content-style pairs and 1,000 total stylizations across the two variants. The content images come from Unsplash and are listed in the project's image manifest. The style images come from a public style-transfer dataset. Links to both resources are provided in Section 5.

We generated 300 precomputed depth masks, with one foreground, midground, and background mask for each content image. These masks were produced using MiDaS and refined through manual threshold adjustment. The reported evaluation therefore reflects a pipeline with manually tuned masks.

3.2 Evaluation Metrics

For each output, we compute complementary metrics that describe structural fidelity, perceptual distance, and pixel-space reconstruction error relative to the content image. We report the Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR), mean squared error (MSE), and mean absolute error (MAE). Higher SSIM and PSNR indicate greater agreement with the content image; lower MSE and MAE indicate smaller pixel differences.

We also compute Learned Perceptual Image Patch Similarity (LPIPS) [4] using the AlexNet backbone, with images rescaled to $[-1, 1]$ as expected by the implementation [3]. Lower LPIPS indicates that the output is perceptually closer to the content image.

To examine the style-content tradeoff, we compute VGG feature-based content and style losses that mirror the baseline objective. We summarize their relative magnitudes using

$$\mathrm{S2C} = \frac{\mathcal{L}_{\text{style}}}{\mathcal{L}_{\text{content}} + \varepsilon}, \tag{7}$$

where ε is a small stabilizing constant. This is a ratio of losses, so it should be interpreted as a diagnostic rather than a direct measure of artistic style strength. It can increase when content loss decreases, when style loss increases, or both.

The `evaluate_stylization` API aggregates results per output and supports optional depth-layer diagnostics. These include a boundary artifact score around mask edges. Because the report does not provide a corresponding baseline boundary score, this diagnostic does not support a direct comparison between the two methods.

Table 1: Aggregate evaluation results. Arrows indicate the preferred direction for similarity to the content image. S2C and the boundary score are diagnostics without an assigned preference.

Metric	Baseline	Depth-aware
SSIM ↑	0.2250	0.4087
PSNR (dB) ↑	10.5144	13.6209
MSE ↓	0.1097	0.0581
MAE ↓	0.2581	0.1749
LPIPS ↓	≈ 0.717	0.6284
Style-to-content ratio	0.00014	0.00022
Boundary artifact score	—	45.079465

The baseline LPIPS value follows the rounded distribution summary in the original report. A separate paragraph gives 0.714; the source results should be checked before reporting a more precise baseline value. The dash indicates an unreported baseline boundary score.

4 Results

4.1 Structural Fidelity and Reconstruction

Table 1 summarizes the aggregate results. We found that the depth-aware method shifts the average output toward higher structural similarity and lower reconstruction error. Mean SSIM increases from 0.2250 to 0.4087, a gain of 0.1837 based on the displayed values. PSNR increases from 10.5144 to 13.6209 dB, a gain of approximately 3.11 dB.

Pixel-space errors show the same trend. MSE decreases by 0.0516, from 0.1097 to 0.0581, and MAE decreases by 0.0832, from 0.2581 to 0.1749. These correspond to reductions of approximately 47.0% and 32.2%, respectively. Figure 1 preserves the aggregate comparison from our original evaluation.

These gains are consistent with the blending formulation. From Equation 5, $Y(p) - C(p) = A(p)[T(p) - C(p)]$, where $0.3 \le A(p) \le$

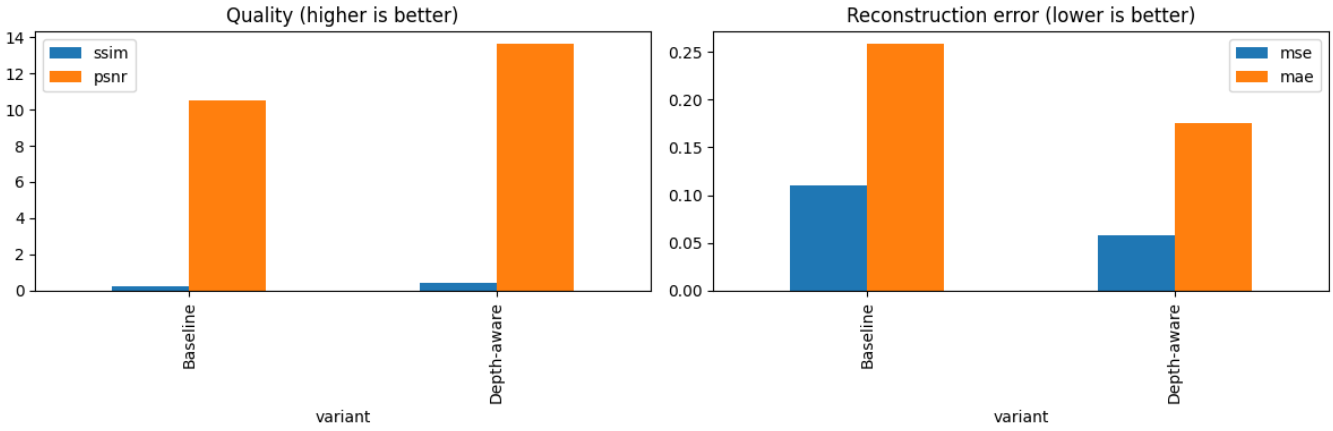


Figure 1: Original aggregate comparison of quality and reconstruction metrics. SSIM and PSNR have different units and scales; their numerical values are reported separately in Table 1.

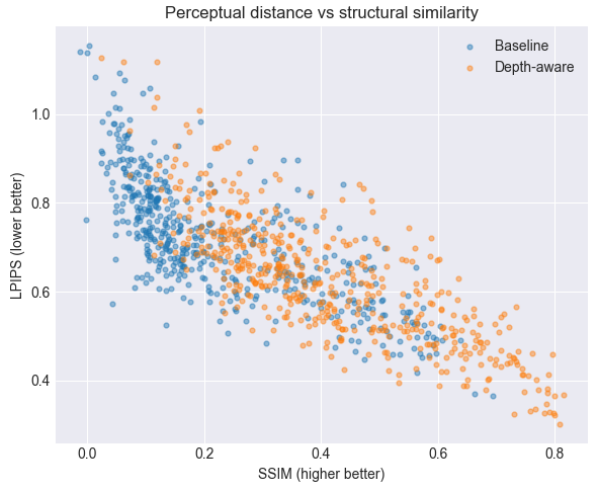


Figure 2: Perceptual distance versus structural similarity for the baseline and depth-aware outputs. Lower LPIPS and higher SSIM indicate greater similarity to the content image.

1. Blending therefore reduces or preserves the magnitude of each pixel's deviation from the content image. This helps explain the lower MSE and MAE, but it also means that reconstruction gains alone do not establish an advantage over a uniformly blended baseline.

4.2 Perceptual Distance and Variation

The reported LPIPS results also indicate that depth-aware outputs remain closer to the original content. The distribution summary gives baseline LPIPS of 0.717 ± 0.134 and depth-aware LPIPS of 0.628 ± 0.140, where the values after $\pm$ are standard deviations. The corresponding SSIM summaries are 0.225 ± 0.159 for the baseline and 0.409 ± 0.178 for the depth-aware method.

The SSIM-LPIPS scatter in Figure 2 shows a strong negative association. Across both variants, the reported Pearson correlation is $r = -0.789$. Within each variant, the correlations remain negative: $r_b = -0.736$ for the baseline and $r_d = -0.796$ for the depth-aware method. Outputs that preserve more structure therefore also tend to be perceptually closer to the content. These are descriptive associations; we do not report statistical significance tests or claim that every individual image improves.

4.3 Style-Content Tradeoff and Visual Examples

The style-to-content ratio increases from approximately 1.40×10^{-4} to 2.22×10^{-4} in the original narrative, while the table rounds the latter to 0.00022. This change shows that the relative balance of the two losses shifts after blending. It does not, by itself, establish that stylization strength is preserved: the denominator can fall as the output moves closer to the content, and a larger style loss can indicate a poorer match to the style target.

The visual examples provide a more direct view of the intended effect. In Figure 3, depth-aware blending preserves more of the elephant's facial and body details while leaving strong color and texture in the surrounding scene. In Figure 4, the house and foreground remain closer to their original appearance while the sky retains visible stylization. These examples illustrate how the same baseline result can be redistributed across depth layers.

5 Implementation Details

We implemented the project in Python 3.10+ using PyTorch, Torchvision, NumPy, Pillow, OpenCV, Matplotlib, and Ipywidgets. MiDaS supplies depth estimates, VGG19 supplies content and style features, and the LPIPS package supplies perceptual comparisons. The implementation is organized into the following components:

- `style_transfer.py` implements the baseline normalization, content loss, and style loss modules. It constructs the optimization graph, runs L-BFGS, and exposes a high-level stylization API.



Figure 3: Elephant example, shown from left to right as the content image, style image, baseline transfer, and depth-aware transfer. Foreground details remain more visible after blending. For this example, SSIM increases from 0.0974 to 0.3701 and LPIPS decreases from 0.7221 to 0.5915. Images and values are retained from the original report.

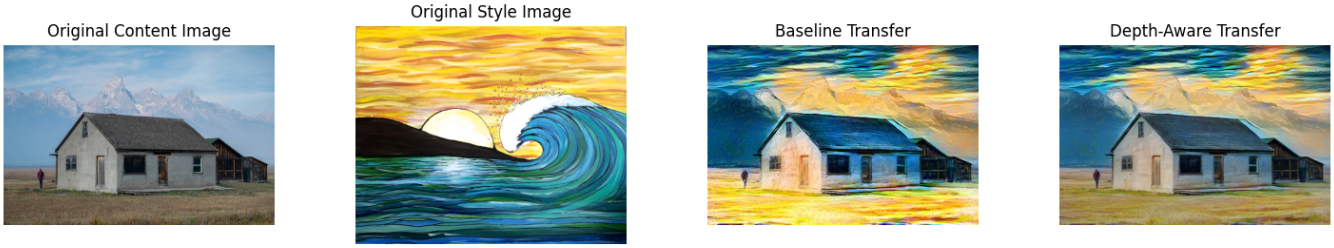


Figure 4: House example in the same order as Figure 3. Depth-aware blending preserves more of the house and ground while retaining a stylized sky. SSIM increases from 0.2330 to 0.4376 and LPIPS decreases from 0.7178 to 0.6447. Images and values are retained from the original report.

- `depth_aware_style_transfer.py` handles depth estimation, soft mask generation, precomputed mask loading, and layer blending.
- `depth_gui.py` provides a Matplotlib interface for threshold adjustment with real-time previews, fullscreen mode, and automatic saving.
- `eval.py` computes structural, perceptual, pixel-space, VGG loss, and boundary diagnostics.
- `demo.ipynb` demonstrates the full pipeline and batch processing. `results_viewer.ipynb` provides a widget-based results browser.

The content-image manifest is available in the project repository[1]. The style-image collection is available in the style-transfer dataset repository[2]. The 300 manually refined masks allow the same content-image segmentation to be reused across styles.

6 Challenges, Limitations, and Future Work

The primary challenge was combining the baseline style transfer pipeline with a depth-based compositing strategy. We needed to interpret the content and style representations, produce useful spatial layers from relative depth, and prevent those layers from creating distracting transitions. Gaussian feathering and per-pixel normalization helped address the transition problem.

Depth estimation remained a practical limitation. Some scenes contained regions that were not separated appropriately by the initial thresholds. We developed the interactive GUI to refine these masks, and this process required several hours of manual tuning. The method therefore depends both on the estimated depth map and on human choices about layer boundaries. Feathering can soften an incorrect boundary, but it cannot correct the underlying depth assignment.

Our evaluation also has limits. The reported comparisons use the fully stylized baseline, while the depth-aware outputs explicitly retain original content. A useful next experiment would compare against uniform blending at matched average strength, followed by ablations of the masks, thresholds, and feathering. This would help isolate the contribution of depth from the contribution of blending itself. Separate foreground and background measurements, along with a user study, would provide stronger evidence about perceived depth, artistic quality, and style preservation.

The current report does not specify all settings needed for exact reproduction, including the MiDaS checkpoint, image resolution, optimization iteration count, loss weights, and Gaussian standard

[1] https://github.com/TheoXiong7/CS445_FinalProject
[2] https://github.com/victorkitov/style-transfer-dataset

deviation. It also does not report hardware, runtime, or raw per-image measurements. Future experiments should record these settings and resolve the conflicting baseline LPIPS summaries. For video or augmented reality, temporal consistency would be an additional requirement, since independently estimated masks could change between frames.

7 Conclusion

Our project introduces depth awareness to neural style transfer through a simple compositing stage. We generate one stylized image, estimate relative depth, and use soft spatial masks to apply lighter stylization to the foreground and stronger stylization to the background. The reported experiments show higher average structural similarity and lower pixel-space and perceptual distances to the original content, while the examples retain visible artistic effects in distant regions.

Through this work, we explored how geometric information can guide image stylization without changing the baseline neural optimization. The results support depth-dependent blending as a useful tool for controlling where stylization appears. Further comparisons against matched uniform blending and evaluations of artistic preference would clarify how much of the improvement comes specifically from depth awareness.

References

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